

SYNOPSIS CHAMPIONSHIP PROJECT ABSTRACT

The Abstract is a required part of your project. Bring 1 copy at Project Check-in.

Your abstract should be written after you finish the research and experimentation and should include:

- **Project number**, title and the full name of every team member
- Purpose of the project
- Procedure
- Observations/data/results
- Conclusions
- 250 words or less (to fit on one page)
- Use a black font size 10 to 12

Project Number: A31

Project Title: Vision-Language Navigation over Semantically Annotated 3D Maps

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Autonomous outdoor vehicles like Waymo have transformed transportation, yet indoor spaces such as homes, warehouses, and retail floors remain largely unaddressed by similar robotic navigation systems. We present a navigation system bringing the full-stack self-driving paradigm indoors: converting 3D mesh environments into semantically-annotated navigable maps and layering a Vision-Language Model (VLM) decision engine atop a high-performance classical planner. Our system starts by processing a raw 3D environment mesh, utilizing Recast to generate a NavMesh representing contiguous walkable regions. Raw sensor input (LiDAR, RGB cameras, and an IMU) are fused using a Kalman Filter to construct an occupancy grid. Once this geometric map is developed, we populate the scene with semantic annotations by running an exploration pass. We first use a VLM to list relevant scene features. We then use CLIP-based segmentation to identify object regions in 2D, which are back-projected into the 3D map. With this context, we layer a VLM to interpret semantically rich instructions such as "find a soft place to sit with a nearby surface for a laptop" and select a corresponding waypoint. Navigation to a waypoint is delegated to a custom classical planner, which uses A* for shortest path waypoints over the NavMesh and a ray-marching local planner for safe obstacle clearance. On Habitat-Lab ObjectNav and PointNav we match state-of-the-art Success and SPL under map-based evaluation while outperforming some existing methods on more semantically complex queries. These results pose our pipeline as a potentially viable path towards enhancing autonomous indoor navigation.